1 Methods

1.1 Core Idea

1.1.1 Setup

Consider a logistic regression, with missing features in the observation.
Let $x := (x^o, x^m) \in \mathbb{R}^d$. Then the logistic regression is defined as

$$p(y \mid x) := \sigma(w^T x), \qquad w \in \mathbb{R}^d.$$

Additionally we have a PC that models the density $p(x)$.

1.1.2 Goal

Compute the prediction in a probabilistically principled manner using

$$p(y \mid x^o) := \mathbb{E}_{p(x^m|x^o)} \left[\sigma\left(w^T x\right)\right].$$

If it were not for the non-linear sigmoid function σ, we would be able to calculate this in a smooth and decomposable PC, by pulling the Expectation -which is linear- into the leaf nodes. To tackle this we propose to use two tricks.

1.1.3 Trick 1: Pólya-Gamma

To replace the sigmoid function we use the PG-Sigmoid identity, shown in [1], that states:

$$\sigma(z) = \frac{1}{2} \exp\left(\frac{1}{2} z\right) \ \mathbb{E}_{\gamma \sim \mathrm{PG}(1,0)} \left[\exp\left(-\frac{1}{2}\gamma z^2\right)\right]$$

for $z \in \mathbb{R}$.

If we set $z := w^T x$, and insert into our expectation, we get:

$$
\begin{aligned}
p(y \mid x^o) &= \mathbb{E}_{p(x^m|x^o)}\left[\sigma(z)\right] \\
&= \mathbb{E}_{p(x^m|x^o)}\left[\frac{1}{2}\exp\left(\frac{1}{2}z\right)\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\left[\exp\left(-\frac{1}{2}\gamma z^2\right)\right]\right] \\
&= \frac{1}{2}\mathbb{E}_{p(x^m|x^o)}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\left[\exp\left(\frac{1}{2}z-\frac{1}{2}\gamma z^2\right)\right]
\end{aligned}
$$

Now, we have a term without a sigmoid, but the x still appears in a quadratic form.

1.1.4 Trick 2: Hubbard-Stratonovich (HS)

To get rid of the quadratic form, we apply the Hubbard–Stratonovich identity [2, 3], that states:

$$
\exp\left(-\tfrac{1}{2}ah^2\right) = \sqrt{\frac{1}{2\pi a}}\int_{-\infty}^{\infty}\exp\left(-\frac{u^2}{2a}-ihu\right)\mathrm{d}u, \tag{1.1}
$$

Since $\frac{1}{\sqrt{2\pi a}}\exp\left(-\frac{u^2}{2a}\right)$ is the density of $\mathcal{N}(0,a)$, the integral is exactly the expectation of e^{-ihu} over $u\sim\mathcal{N}(0,a)$:

$$
\exp\left(-\tfrac{1}{2}ah^2\right) = \mathbb{E}_{u\sim\mathcal{N}(0,\,a)}\left[e^{-ihu}\right]. \tag{1.2}
$$

In plain words, it makes a quadratic term linear at the cost of introducing a helper variable that we need to integrate over.

Currently, in our term, x appears quadratically as well as linearly. To be able to apply HS, we complete the square (again using $z = w^T x$):

$$
\begin{aligned}
\frac{1}{2}z-\frac{1}{2}\gamma z^2 &= -\frac{1}{2}\gamma\left(z^2-\frac{z}{\gamma}\right) \\
&= -\frac{1}{2}\gamma\left(\left(z-\frac{1}{2\gamma}\right)^2-\frac{1}{4\gamma^2}\right) \\
&= -\frac{1}{2}\gamma\left(z-\frac{1}{2\gamma}\right)^2+\frac{1}{2}\gamma\cdot\frac{1}{4\gamma^2} \\
&= -\frac{1}{2}\gamma\left(z-\frac{1}{2\gamma}\right)^2+\frac{1}{8\gamma}.
\end{aligned} \tag{1.3}
$$

Now we apply HS with $a = \gamma$ and $h = z - \frac{1}{2\gamma}$:

$$\begin{aligned}
\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right) &= \exp\left(\frac{1}{8\gamma}\right)\exp\left(-\frac{1}{2}\gamma\left(z - \frac{1}{2\gamma}\right)^2\right) \\
&= \exp\left(\frac{1}{8\gamma}\right)\sqrt{\frac{1}{2\pi\gamma}}\int_{-\infty}^{\infty}\exp\left(-\frac{u^2}{2\gamma} - i\left(z - \frac{1}{2\gamma}\right)u\right)\mathrm{d}u \\
&= \exp\left(\frac{1}{8\gamma}\right)\int_{-\infty}^{\infty}\underbrace{\frac{1}{\sqrt{2\pi\gamma}}\exp\left(-\frac{u^2}{2\gamma}\right)}_{\mathcal{N}(u;0,\gamma)}\exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right)\mathrm{d}u \\
&= \exp\left(\frac{1}{8\gamma}\right)\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(-i\left(z - \frac{1}{2\gamma}\right)u\right)\right] \\
&= \exp\left(\frac{1}{8\gamma}\right)\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(\frac{iu}{2\gamma} - izu\right)\right] \qquad (1.4)
\end{aligned}$$

Substituting back into $p(y \mid x^o)$, we get:

$$\begin{aligned}
p(y \mid x^o) &= \frac{1}{2}\mathbb{E}_{p(x^m|x^o)}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\left[\exp\left(\frac{1}{2}z - \frac{1}{2}\gamma z^2\right)\right] \\
&= \frac{1}{2}\mathbb{E}_{p(x^m|x^o)}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\left[\exp\left(\frac{1}{8\gamma}\right)\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(\frac{iu}{2\gamma} - izu\right)\right]\right] \\
&= \frac{1}{2}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(\frac{1+4iu}{8\gamma}\right)\mathbb{E}_{p(x^m|x^o)}\left[\exp\left(-izu\right)\right]\right]
\end{aligned}$$

1.1.5 Characteristic Function

The characteristic function of a random variable X is defined as $\phi_X(s) = \mathbb{E}\left[\exp(isX)\right]$. The innermost expectation is only over x^m, with x^o fixed, so we split $z = w^T x$ into its observed and missing parts:

$$z = w^T x = w_o^T x^o + w_m^T x^m.$$

The x^o part is constant under $\mathbb{E}_{p(x^m|x^o)}$ and factors out:

$$\begin{aligned}\mathbb{E}_{p(x^m|x^o)}\left[\exp(-izu)\right] &= \mathbb{E}_{p(x^m|x^o)}\left[\exp\left(-i(w_o^T x^o + w_m^T x^m)u\right)\right] \\ &= \exp\left(-iw_o^T x^o u\right)\cdot \mathbb{E}_{p(x^m|x^o)}\left[\exp\left(-iw_m^T x^m u\right)\right] \\ &= \exp\left(-iw_o^T x^o u\right)\cdot \phi_{x^m|x^o}(-uw_m)\end{aligned}$$

So the input to the characteristic function is $s := -uw_m$.

1.1.6 Full Formula

Plugging the result of the previous subsection back into the outer expression for $p(y \mid x^o)$, we obtain:

$$\begin{aligned}p(y \mid x^o) &= \frac{1}{2}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(\frac{1+4iu}{8\gamma}\right)\cdot \exp\left(-iw_o^T x^o u\right)\cdot \phi_{x^m|x^o}(-uw_m)\right] \\ &= \frac{1}{2}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(\frac{1}{8\gamma}+\frac{iu}{2\gamma}-iw_o^T x^o u\right)\cdot \phi_{x^m|x^o}(-uw_m)\right] \\ &= \frac{1}{2}\mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}\left[\exp\left(\frac{1}{8\gamma}+iu\left(\frac{1}{2\gamma}-w_o^T x^o\right)\right)\cdot \phi_{x^m|x^o}(-uw_m)\right]\end{aligned}$$

We propose this as a final expression that we can tractably evaluate.

1.2 Implementation

Two ingredients are still abstract in the formula above: the conditional characteristic function $\phi_{x^m|x^o}$, which depends on the density model, and the inner $\mathcal{N}(0,\gamma)$ expectation, which needs to be approximated. This section makes both concrete.

1.2.1 Conditional CF for a GMM

For a Gaussian mixture model $p(x) = \sum_{k=1}^K \pi_k\,\mathcal{N}(x;\mu_k,\Sigma_k)$ with diagonal covariances $\Sigma_k = \mathrm{diag}(\sigma_k^2)$, the conditional density $p(x^m \mid x^o)$ is again a mixture, with the same component shapes restricted to the missing coordinates and reweighted by the posterior responsibility of each component given x^o:

$$\tilde{\pi}_k(x^o) = \frac{\pi_k\,\mathcal{N}(x^o;\mu_k^o,\Sigma_k^o)}{\sum_{j=1}^K \pi_j\,\mathcal{N}(x^o;\mu_j^o,\Sigma_j^o)}.$$

Since the characteristic function of $\mathcal{N}(\mu, \Sigma)$ at s is $\exp\left(is^T\mu - \frac{1}{2}s^T\Sigma s\right)$, and the CF of a mixture is the same convex combination of component CFs, we get a closed-form expression:

$$\phi_{x^m|x^o}(s) = \sum_{k=1}^{K} \tilde{\pi}_k(x^o) \exp\left(is^T\mu_k^m - \tfrac{1}{2}s^T\Sigma_k^m s\right),$$

where μ_k^m, Σ_k^m are the component means and covariances restricted to the missing coordinates.

1.2.2 Gauss–Hermite Quadrature

The inner Gaussian expectation is amenable to Gauss–Hermite quadrature, with deterministic nodes $\{t_i\}_{i=1}^N$ and weights $\{\omega_i\}_{i=1}^N$ such that

$$\int_{-\infty}^{\infty} e^{-t^2} g(t)\,\mathrm{d}t \approx \sum_{i=1}^{N} \omega_i\, g(t_i)$$

To use this we need to bring the $\mathcal{N}(0, \gamma)$ expectation into the form $\int e^{-t^2} g(t)\,\mathrm{d}t$. Writing the expectation out as an integral against the Gaussian density gives

$$\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}[f(u)] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} \exp\left(-\frac{u^2}{2\gamma}\right) f(u)\,\mathrm{d}u.$$

Apply the substitution $u = \sqrt{2\gamma}\,t$. It changes three pieces of the integrand:

- the differential: $\mathrm{d}u = \sqrt{2\gamma}\,\mathrm{d}t$,
- the Gaussian exponent: $\dfrac{u^2}{2\gamma} = \dfrac{(\sqrt{2\gamma}\,t)^2}{2\gamma} = \dfrac{2\gamma\,t^2}{2\gamma} = t^2$,
- the argument of f: $f(u) = f(\sqrt{2\gamma}\,t)$,

and the integration limits are unchanged since $\sqrt{2\gamma} > 0$. Plugging all three substitutions into the integral:

$$\begin{aligned}
\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}[f(u)] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\gamma}} e^{-t^2} f\left(\sqrt{2\gamma}\,t\right) \sqrt{2\gamma}\,\mathrm{d}t \\
&= \frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} \int_{-\infty}^{\infty} e^{-t^2} f\left(\sqrt{2\gamma}\,t\right) \mathrm{d}t.
\end{aligned}$$

The constant prefactor simplifies to

$$\frac{\sqrt{2\gamma}}{\sqrt{2\pi\gamma}} = \sqrt{\frac{2\gamma}{2\pi\gamma}} = \sqrt{\frac{1}{\pi}} = \frac{1}{\sqrt{\pi}},$$

so the Gaussian expectation becomes

$$\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}[f(u)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2} f\left(\sqrt{2\gamma}\, t\right) \mathrm{d}t.$$

Which we then can apply by

$$\mathbb{E}_{u\sim\mathcal{N}(0,\gamma)}[f(u)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^{N} \omega_i\, f\left(\sqrt{2\gamma}\, t_i\right).$$

Plugging this into the inner expectation of our final formula, with

$$f(u) = \exp\left(\frac{1}{8\gamma} + iu\left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}(-uw_m)$$

gives

$$p(y \mid x^o) \approx \frac{1}{2\sqrt{\pi}} \mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)} \left[\sum_{i=1}^{N} \omega_i\, \exp\left(\frac{1}{8\gamma} + i\sqrt{2\gamma}\, t_i \left(\frac{1}{2\gamma} - w_o^T x^o\right)\right) \phi_{x^m|x^o}\left(-\sqrt{2\gamma}\, t_i\, w_m\right)\right]$$

1.2.3 Approximating the PG integral

We also want don't want to sample the outer expectation over $\gamma \sim \mathrm{PG}(1,0)$ and construct a set of fixed nodes and weights that "approximate" the PG(1,0) measure best. From what we found, there are multiple ways to do this.

Golub–Welsch construction

One is the *Golub–Welsch algorithm* [4], which produces n-point quadrature nodes and weights from the first $2n$ moments of any positive measure.

The Laplace transform of $\mathrm{PG}(1,0)$ is known in closed form:

$$\mathcal{L}(t) := \mathbb{E}_{\gamma\sim\mathrm{PG}(1,0)}\left[e^{-t\gamma}\right] = \operatorname{sech}\left(\sqrt{\tfrac{t}{2}}\right) = \frac{1}{\cosh\left(\sqrt{t/2}\right)}.$$

The raw moments follow from derivatives at zero:

$$\mu_k := \mathbb{E}[\gamma^k] = (-1)^k\, \mathcal{L}^{(k)}(0),$$

which can be evaluated numerically to arbitrary precision.

Given $\mu_0, \dots, \mu_{2n-1}$, the algorithm goes as follows. Form the $n \times n$ Hankel moment matrices

$$H_{ij} = \mu_{i+j}, \qquad H'_{ij} = \mu_{i+j+1}, \quad i, j = 0, \dots, n-1.$$

Compute the Cholesky factor $H = R^\top R$ and the Jacobi (tridiagonal) matrix

$$J = R^{-\top} H' R^{-1}, \quad \text{symmetrised as } J \leftarrow \tfrac{1}{2}(J + J^\top).$$

Eigendecompose $J = V \Lambda V^\top$. Then the quadrature nodes and weights are

$$\gamma_j = \lambda_j, \qquad w_j = \mu_0 \, v_{0j}^2, \quad j = 1, \dots, n,$$

where λ_j are the eigenvalues and v_{0j} is the first component of the j-th eigenvector. The resulting rule integrates polynomials in γ exactly up to degree $2n - 1$.

Nonnegative least-squares construction

The Golub–Welsch rule matches the *moments* of $\mathrm{PG}(1, 0)$, i.e. the Taylor coefficients of $\mathcal{L}$ at $t = 0$, so its accuracy is concentrated near $z = 0$ and decays as $|z|$ grows. An alternative is to match the Laplace transform directly across the range of z.

Fix M candidate support locations $\{\gamma_j\}_{j=1}^M$ log-spaced in $[\gamma_{\min}, \gamma_{\max}]$, and $R \gg M$ fitting points $\{z_r\}_{r=1}^R$ linearly spaced in $[0, z_{\max}]$. Define

$$A_{rj} = e^{-z_r \gamma_j}, \qquad b_r = \mathcal{L}(z_r) = \operatorname{sech}\left(\sqrt{z_r/2}\right),$$

and solve the nonnegative least-squares problem

$$\min_{w \geq 0} \; \|Aw - b\|_2^2.$$

The solution w is then normalized to sum to one, and support points with $w_j < \varepsilon$ are eliminated. The remaining pairs (γ_j, w_j) serve as quadrature nodes and weights for the outer PG integral.

Fully deterministic formula.

Combining PG nodes $(\gamma_j, w_j)_{j=1}^M$ with Gauss–Hermite nodes $(t_i, \omega_i)_{i=1}^N$ yields the deterministic approximation

$$p(y \mid x^o) \approx \frac{1}{2\sqrt{\pi}} \sum_{j=1}^M w_j \sum_{i=1}^N \omega_i \, \exp\left(\frac{1}{8\gamma_j} + i\sqrt{2\gamma_j}\, t_i \left(\frac{1}{2\gamma_j} - w_o^T x^o \right) \right) \phi_{x^m|x^o}\left(-\sqrt{2\gamma_j}\, t_i \, w_m \right).$$

1.2.4 Algorithms

Putting the pieces together, we can define three variants of our method, depending on how we want to approximate the integrals. A full Monte Carlo version samples both γ_j and u_i; a Gauss–Hermite version samples only γ_j and uses quadrature for u_i; and a fully deterministic version uses quadrature for both.

Full Monte Carlo

Below is an algorithm for the simplest - fully Monte Carlo - variant.

Algorithm 1 PCPG: expected sigmoid under a GMM, full Monte Carlo

Input: LR weights w, partial observation x^o, GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, # PG samples M, # Gaussian samples N

Split $w = (w_o, w_m)$ and $\mu_k = (\mu_k^o, \mu_k^m)$, $\Sigma_k = (\Sigma_k^o, \Sigma_k^m)$ by the observed/missing mask

$\tilde{\pi}_k \leftarrow \dfrac{\pi_k \, \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \, \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \dots, K$

$c \leftarrow w_o^T x^o$

$S \leftarrow 0$

for $j = 1, \dots, M$ **do**

  Sample $\gamma_j \sim \mathrm{PG}(1, 0)$

  for $i = 1, \dots, N$ **do**

    Sample $u \sim \mathcal{N}(0, \gamma_j)$

    $s \leftarrow -u \, w_m$

    $\phi \leftarrow \sum_{k=1}^K \tilde{\pi}_k \exp\Big(i s^T \mu_k^m - \frac{1}{2} s^T \Sigma_k^m s\Big)$

    $\text{factor} \leftarrow \exp\left(\dfrac{1}{8\gamma_j} + iu\left(\dfrac{1}{2\gamma_j} - c\right)\right)$

    $S \leftarrow S + \text{factor} \cdot \phi$

  end for

end for

return $\dfrac{1}{2\,M\,N} \cdot \mathrm{Re}(S)$

Unified version

Since all variant share the same core structure, we don't include separate algorithms for all, but but instead give a unified version that takes the outer and inner nodes/weights as input. Table 1.1 summarises the differences; Algorithm 2 gives the unified procedure.

Variant	**Outer** (γ_j, w_j)	**Inner** $(u_i, \omega_i \mid \gamma_j)$	**Norm** Z
PCPG-MC	M samples $\mathrm{PG}(1,0)$, $w_j{=}1$	N samples $\mathcal{N}(0,\gamma_j)$, $\omega_i{=}1$	$\frac{1}{2MN}$
PCPG-GH	M samples $\mathrm{PG}(1,0)$, $w_j{=}1$	Hermite: $u_i{=}\sqrt{2\gamma_j}\, t_i$	$\frac{1}{2\sqrt{\pi}M}$
PCPG-Quad	GW nodes/weights	Hermite: $u_i{=}\sqrt{2\gamma_j}\, t_i$	$\frac{1}{2\sqrt{\pi}}$

Table 1.1: The three PCPG variants. GW = Golub–Welsch (§1.2.3); Hermite nodes (t_i, ω_i) from `hermgauss`.

Algorithm 2 PCPG (unified)

Input: LR weights w, partial observation x^o, GMM parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, outer iterates $(\gamma_j, w_j)_{j=1}^M$, inner iterates $(u_i, \omega_i)_{i=1}^N$ (per γ_j), normalisation Z

Split $w = (w_o, w_m)$ by the observed/missing mask

$\tilde{\pi}_k \leftarrow \dfrac{\pi_k \mathcal{N}(x^o; \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x^o; \mu_j^o, \Sigma_j^o)}$ for $k = 1, \ldots, K$

$c \leftarrow w_o^T x^o, \quad S \leftarrow 0$

for $j = 1, \ldots, M$ **do**

  for $i = 1, \ldots, N$ **do**

    $s \leftarrow -u_i\, w_m$

    $\phi \leftarrow \sum_k \tilde{\pi}_k \exp\Big(i s^T \mu_k^m - \frac{1}{2} s^T \Sigma_k^m s\Big)$

    $\text{factor} \leftarrow \exp\left(\dfrac{1}{8\gamma_j} + i u_i \left(\dfrac{1}{2\gamma_j} - c\right)\right)$

    $S \leftarrow S + w_j\, \omega_i\, \text{factor} \cdot \phi$

  end for

end for

return $Z \cdot \mathrm{Re}(S)$

Of course one could also add a forth variant that uses quadrature for the PG integral but sampling for the inner Gaussian, but we don't include it here since it doesn't perform well in practice. The variance of the inner Gaussian sampling is very high which makes the quadrature perform much better.

2 Discussion

2.1 Variance Analysis

2.1.1 Conditional Sampling MC Estimator

The MC estimator draws N samples $x^{m(i)} \sim p(x^m \mid x^o)$ and returns

$$\widehat{p}_{\mathrm{MC}}(N) = \frac{1}{N} \sum_{i=1}^{N} \sigma\Big(w^T x^{(i)}\Big) .$$

Since $\sigma(\cdot) \in (0,1)$, the per-sample variance $\sigma^2_{\mathrm{MC}} := \mathrm{Var}_{p(x^m|x^o)}\Big[\sigma(w^T x)\Big]$ is bounded by $1/4$, giving

$$\mathrm{Var}[\widehat{p}_{\mathrm{MC}}(N)] = \frac{\sigma^2_{\mathrm{MC}}}{N} \;\leq\; \frac{1}{4N}.$$

The $1/N$ factor follows from the standard variance-of-a-sample-mean identity: since the draws are i.i.d., the variance of their average is the per-sample variance divided by N. The bound $\sigma^2_{\mathrm{MC}} \leq 1/4$ holds because $\sigma(\cdot) \in (0,1)$: for any $[0,1]$-valued random variable Y,

$$\mathrm{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \leq \mathbb{E}[Y] - \mathbb{E}[Y]^2 = \mathbb{E}[Y](1 - \mathbb{E}[Y]) \leq \tfrac{1}{4},$$

where the first inequality uses $Y^2 \leq Y$ on $[0,1]$ and the second uses $p(1-p) \leq 1/4$ for all $p \in [0,1]$, with equality at $p = 1/2$.

2.1.2 PCPG-MC Estimator

The PCPG-MC estimator with M outer (PG) and L inner (Gaussian) samples returns

$$\widehat{p}_{\mathrm{PCPG}}(M, L) = \frac{1}{2ML} \sum_{j=1}^{M} \sum_{i=1}^{L} \mathrm{Re}[h(\gamma_j, u_{ij})] ,$$

where $h(\gamma, u) := \exp\Big(\frac{1}{8\gamma} + iu\Big(\frac{1}{2\gamma} - c\Big)\Big) \cdot \phi_{x^m|x^o}(-u w_m)$, $c = w_o^T x^o$, and $u_{ij} \sim \mathcal{N}(0, \gamma_j)$.

Output range

A single sample (M=L=1) returns $\frac{1}{2}\mathrm{Re}[h(\gamma, u)]$.

The characteristic function $\phi(s) = \mathbb{E}[e^{isx}]$ satisfies $|\phi(s)| \leq 1$ for all s.

Since $|h(\gamma, u)| = e^{1/(8\gamma)} \cdot |\phi(-uw_m)| \leq e^{1/(8\gamma)}$, the value lies in $\left[-\frac{1}{2}e^{1/(8\gamma)},\ +\frac{1}{2}e^{1/(8\gamma)}\right]$.

Because $\gamma \sim \mathrm{PG}(1, 0)$ can be arbitrarily small, $e^{1/(8\gamma)}$ is unbounded, so individual PCPG samples are not confined to $[0, 1]$ and can be negative or exceed 1.

By contrast, every individual MC sample lies in $(0, 1)$.

Variance

The PCPG-MC estimator has two nested levels of randomness.

1. **Outer level**: M independent draws $\gamma_j \sim \mathrm{PG}(1, 0)$.
2. **Inner level**: for each j, L independent draws $u_{ij} \sim \mathcal{N}(0, \gamma_j)$.

Let $Z = \frac{1}{2}\mathrm{Re}[h(\gamma, u)]$ denote a single "raw" sample. The estimator is the mean of ML such values, grouped so that each outer sample γ_j drives a block of L inner values.

Step 1: variance within one outer block. Fix γ_j. The L inner samples u_{ij} are i.i.d. $\mathcal{N}(0, \gamma_j)$, so the block average $\bar{Z}_j := \frac{1}{L}\sum_{i=1}^{L} Z_{ij}$ has conditional variance $\sigma^2_{A,\gamma_j}/L$, where $\sigma^2_{A,\gamma} := \mathrm{Var}_{u|\gamma}[Z \mid \gamma]$.

Step 2: law of total variance for $\bar{Z}_j$. The M blocks are independent, so it suffices to study a single block. Applying the law of total variance across the two levels of randomness:

$$\mathrm{Var}[\bar{Z}_j] = \underbrace{\mathbb{E}_\gamma\Big[\mathrm{Var}[\bar{Z}_j \mid \gamma_j]\Big]}_{\text{avg. inner variance}} + \underbrace{\mathrm{Var}_\gamma\Big[\mathbb{E}[\bar{Z}_j \mid \gamma_j]\Big]}_{\text{outer PG variance}} = \frac{\sigma_A^2}{L} + \sigma_B^2,$$

where

$$\sigma_A^2 := \mathbb{E}_\gamma\Big[\sigma^2_{A,\gamma}\Big] \qquad \text{and} \qquad \sigma_B^2 := \mathrm{Var}_\gamma\Big[\mathbb{E}_{u|\gamma}[Z \mid \gamma]\Big]\,.$$

σ_A^2 measures how much the inner Gaussian integral fluctuates for a typical PG draw. σ_B^2 measures how much the PG draw γ itself shifts the result.

Step 3: overall variance. The estimator is $\widehat{p} = \frac{1}{M}\sum_{j=1}^{M} \bar{Z}_j$ with M i.i.d. blocks, so

$$\mathrm{Var}[\widehat{p}_{\mathrm{PCPG}}(M,L)] = \frac{1}{M}\,\mathrm{Var}[\bar{Z}_j] = \frac{\sigma_A^2}{ML} + \frac{\sigma_B^2}{M}.$$

Step 4: fixed-ratio budget. Writing the total budget as $N = ML$ and the inner/outer ratio as $r = L/M$ (so $M = \sqrt{N/r}$):

$$\mathrm{Var}[\widehat{p}_{\mathrm{PCPG}}(M,L)] = \frac{\sigma_A^2}{N} + \frac{\sigma_B^2\sqrt{r}}{\sqrt{N}}.$$

The first term decays as $O(1/N)$, the same rate as MC. The second term decays only as $O(1/\sqrt{N})$, which is *slower*. For large N the second term dominates, so PCPG-MC will have higher variance than MC whenever $\sigma_B^2 > 0$.

Why σ_A^2 can be large. Because $|h(\gamma,u)| = e^{1/(8\gamma)}|\phi|$, individual samples explode when γ is small. The distribution $\mathrm{PG}(1,0)$ places non-negligible mass near zero, so the inner variance $\sigma_{A,\gamma}^2$ diverges as $\gamma \to 0$: formally $\sigma_A^2 = \infty$. In practice the contribution from very small γ is finite but can be large, requiring a moderately large L to suppress the inner noise.

Optimal budget split. Looking at the variance formula from Step 3,

$$\mathrm{Var} = \frac{\sigma_A^2}{ML} + \frac{\sigma_B^2}{M},$$

the two terms respond to the budget differently. The first term shrinks when L is large (many inner samples per PG draw). The second term shrinks when M is large (many PG draws). Since $N = ML$ is fixed, making L larger forces M smaller and vice versa — you cannot reduce both terms simultaneously.

Substituting $M = N/L$ and minimising over L at fixed N gives the optimal split:

$$L^* = \sqrt{\frac{N\,\sigma_A^2}{\sigma_B^2}}, \qquad M^* = \sqrt{\frac{N\,\sigma_B^2}{\sigma_A^2}}, \qquad r^* := \frac{L^*}{M^*} = \frac{\sigma_A^2}{\sigma_B^2}.$$

In words: use more inner samples than outer samples if and only if $\sigma_A^2 > \sigma_B^2$. Because the inner variance is typically much larger (the $e^{1/(8\gamma)}$ blowup), the recommended ratio is $r = 4$, meaning four inner samples per PG draw. At $N = 1000$ and $r = 4$ this gives $M^* \approx 16$ outer draws and $L^* \approx 63$ inner samples per draw; equal splitting ($M = L = 32$) would leave the large inner variance insufficiently averaged.

The minimum variance at the optimal split is $2\sqrt{\sigma_A^2\sigma_B^2}/\sqrt{N}$, which is still $O(1/\sqrt{N})$. No choice of r removes the $O(1/\sqrt{N})$ scaling as long as $\sigma_B^2 > 0$.

When does σ_B^2 become small? Recall that $\sigma_B^2 = \mathrm{Var}_\gamma\Big[\mathbb{E}_{u|\gamma}[Z \mid \gamma]\Big]$: it measures how much the value of the inner integral shifts as γ changes. For a GMM with diagonal covariance, the CF of the missing features satisfies

$$\Big|\phi_{x^m|x^o}(-uw_m)\Big| \;\le\; \exp\!\left(-\frac{u^2}{2}\,D_m\right), \qquad D_m := \sum_{j\in\mathrm{mis}} w_j^2\,\sigma_{kj}^2,$$

where D_m collects the variance of the missing part of $w^T x$. When D_m is large, the CF decays very fast in u: the integrand $h(\gamma, u)$ is nearly zero unless $|u|$ is small. For any fixed γ, the relevant u values are of order $\sqrt{\gamma}$; large D_m means even moderately large γ is enough to zero out the integrand. As a result, the inner expectation $\mathbb{E}_{u|\gamma}[Z \mid \gamma]$ becomes nearly independent of γ — it is close to zero for a wide range of γ values and hardly varies — so $\sigma_B^2 \approx 0$.

Effect of missing ratio (fixed total dimension). Increasing the fraction of missing features (larger d_m, same d) adds more terms to the sum defining D_m, directly increasing it. The CF damps more strongly, σ_B^2 decreases, and PCPG-MC recovers closer to $O(1/N)$ convergence. MC behaves in the opposite way: more missing features make $w^T x$ more random, increasing σ_{MC}^2. The PCPG family therefore *gains* relative to MC as the missing ratio grows.

Effect of total dimension (fixed missing ratio). Increasing d while keeping the missing ratio $\rho = d_m/d$ constant grows both d_m and d_o proportionally. The additional terms in D_m are exactly offset by the fact that logistic-regression weights w_j shrink with larger d (the total discriminative power is spread over more features), so D_m stays roughly constant and the CF damping does not improve. At fixed missing ratio, the rate improvement for PCPG-MC described above does *not* automatically follow from adding more features; what matters is the missing-feature energy D_m, not the total dimension alone.

Comparison.

$$\mathrm{Var}[\widehat{p}_{\mathrm{MC}}(N)] = \frac{\sigma_{\mathrm{MC}}^2}{N} \;\le\; \frac{1}{4N}, \qquad \mathrm{Var}[\widehat{p}_{\mathrm{PCPG}}(M, L)] = \frac{\sigma_A^2}{N} + \frac{\sigma_B^2\sqrt{r}}{\sqrt{N}}.$$

For low-missing-rate settings where σ_B^2 is non-negligible, PCPG-MC is less sample-efficient than MC. The $O(1/\sqrt{N})$ slowdown motivates replacing the inner MC with Gauss-Hermite quadrature (PCPG-GH).

2.1.3 PCPG-GH Estimator

Variance

The PCPG-GH estimator keeps the M outer PG draws $\gamma_j \sim \mathrm{PG}(1,0)$ but replaces each inner MC average with a n_H-point Gauss-Hermite rule. Because the inner quadrature is *deterministic* given γ_j, the inner noise term σ_A^2/L vanishes entirely. The only remaining source of variance is the outer PG draw:

$$\mathrm{Var}[\widehat{p}_{\mathrm{GH}}(M)] = \frac{\sigma_B^2}{M}.$$

The total evaluation budget is $N = M \cdot n_H$ (each outer sample evaluates the CF at n_H frequencies), so

$$\mathrm{Var}[\widehat{p}_{\mathrm{GH}}(M)] = \frac{\sigma_B^2\, n_H}{N},$$

which is $O(1/N)$, matching the MC rate and eliminating the slow $O(1/\sqrt{N})$ term that limits PCPG-MC. The factor n_H is a fixed constant: with $n_H = 20$ nodes the per-budget variance is $20\,\sigma_B^2$, which is favourable whenever σ_B^2 is small compared to $\sigma_{\mathrm{MC}}^2/n_H$.

In short, PCPG-GH is a strictly better choice than PCPG-MC at the same outer budget whenever σ_A^2 is non-negligible, because replacing the inner MC with quadrature costs nothing in outer samples and eliminates the dominant noise source.

2.1.4 PCPG-Quad Estimator

Replacing both integrals with quadrature (Golub–Welsch for the outer PG integral, Gauss–Hermite for the inner) makes the estimator fully deterministic:

$$\mathrm{Var}[\widehat{p}_{\mathrm{Quad}}] = 0.$$

The estimator always returns the same value for the same inputs, and its output is bounded since the PG quadrature nodes γ_j are fixed positive numbers — the factor $e^{1/(8\gamma_j)}$ is a known finite constant for each node. The error is purely from quadrature approximation (bias), not variance.

2.2 Bias Analysis

2.2.1 MC Estimator

MC is unbiased for all $N \geq 1$: sampling from $p(x^m \mid x^o)$ and averaging $\sigma(w^T x)$ is a direct Monte Carlo estimate of the target expectation, so $\mathbb{E}[\widehat{p}_{\mathrm{MC}}(N)] = p(y \mid x^o)$ exactly.

2.2.2 PCPG-MC Estimator

PCPG-MC is also unbiased. Both integrals are approximated by ordinary MC, so by linearity of expectation the estimator equals the true double integral, which is $p(y \mid x^o)$ by the PG-sigmoid and Hubbard–Stratonovich identities.

2.2.3 PCPG-GH Estimator

The outer PG integral is still MC (unbiased). The inner integral is replaced by Gauss–Hermite quadrature, which introduces a small approximation error. For N_H quadrature nodes, GH integrates polynomials of degree up to $2N_H - 1$ exactly, and for smooth (analytic) integrands the error decays super-algebraically in N_H. In our setting the integrand is analytic in u, so the bias is negligible already for small N_H (in practice $N_H \geq 20$ is sufficient).

2.2.4 PCPG-Quad Estimator

Both integrals introduce a quadrature error. The Gauss–Hermite inner error is the same as for PCPG-GH. The Golub–Welsch PG quadrature with M nodes integrates polynomials in γ exactly up to degree $2M - 1$; for smooth functions of γ the error again decays rapidly with M. The maximum reliable order is $M \leq 19$ (limited by numerical precision in the moment computation). At $M = 19$ and $N_H \geq 20$, both errors are numerically negligible and the output closely matches the unbiased MC ground truth.

2.3 Comparison

Table 2.1 summarises the properties of all four estimators.

Table 2.1: Summary of all four estimators. N denotes the total evaluation budget. $r = L/M$ is the inner/outer ratio used by PCPG-MC. n_H is the number of Gauss-Hermite nodes (fixed constant). "CF" means the estimator requires evaluating $\phi_{x^m|x^o}$ rather than drawing samples from $p(x^m \mid x^o)$.

Estimator	Variance	Bias	Rate	Needs
MC	σ^2_{MC}/N	0	$O(1/N)$	samples
PCPG-MC	$\dfrac{\sigma^2_A}{N} + \dfrac{\sigma^2_B\sqrt{r}}{\sqrt{N}}$	0	$O(1/\sqrt{N})$	CF
PCPG-GH	$\sigma^2_B\, n_H/N$	$O(e^{-cn_H})$	$O(1/N)$	CF
PCPG-Quad	0	$O(e^{-cM})$	deterministic	CF

Bias. MC and PCPG-MC are exactly unbiased. PCPG-GH and PCPG-Quad introduce small quadrature errors that are negligible at the parameter choices used ($n_H \geq 20$, $M_{\mathrm{PG}} \leq 19$).

Variance and convergence rate. MC achieves $O(1/N)$ variance, bounded above by $1/(4N)$. PCPG-GH also achieves $O(1/N)$, but the effective constant is $\sigma^2_B\, n_H$: it is preferable to MC when σ^2_B is small, which happens precisely when many features are missing and the CF damps strongly. PCPG-MC is the only estimator with a slower $O(1/\sqrt{N})$ rate; the outer PG variance σ^2_B introduces a $\sigma^2_B\sqrt{r}/\sqrt{N}$ term that persists regardless of how the budget is split between M and L. In the high-missing-data regime $\sigma^2_B \approx 0$ and PCPG-MC recovers $O(1/N)$, but in that regime PCPG-GH is still at least as good. PCPG-Quad has zero variance by construction; its "error" is pure bias from finite-order quadrature, which decays exponentially in the number of nodes and is negligible above $n_H = 20$ and $M_{\mathrm{PG}} = 19$.

When to use which. MC is the simplest and most reliable choice when sampling from $p(x^m \mid x^o)$ is easy and low variance is the priority. PCPG-GH is the natural first upgrade: it eliminates the inner MC noise at no extra outer-sample cost and always achieves $O(1/N)$; use it whenever evaluating the CF is cheaper than drawing samples. PCPG-MC should be preferred over PCPG-GH only when the CF is cheap but computing the Gauss-Hermite nodes is costly, or as a reference implementation to isolate σ^2_A empirically. PCPG-Quad is the right choice when a fully deterministic, reproducible result is required — for example when differentiating through the estimator or when multiple evaluations at the same point must agree exactly.

2.4 Effect of Dimensionality

Two distinct quantities can vary: the number of missing features d_m at fixed total dimension d (equivalently, the missing ratio $\rho = d_m/d$), and the total dimension d itself at a fixed ratio ρ. Their effects are different and we treat them separately. The key quantity that links both estimators is the *missing-feature energy*

$$D_m := \sum_{j\in\text{mis}} w_j^2\,\sigma_{kj}^2,$$

which measures how much variance the uncertain part of $w^T x$ contributes.

Effect of the missing ratio (more missing features, fixed d). As d_m grows at fixed d, more terms enter D_m, so D_m increases.

For *MC*, the uncertain part of the dot product is $w_m^T x^m$, whose variance under $p(x^m \mid x^o)$ is exactly D_m. A larger D_m spreads the distribution of $\sigma(w^T x)$ more widely, pushing $\sigma^2_{\text{MC}} = \text{Var}[\sigma(w^T x)]$ toward its upper bound of $1/4$. *MC variance increases with the missing ratio.*

For *PCPG*, the key observation is what the CF value at a frequency actually means. The CF $\phi_X(s) = \mathbb{E}[e^{isX}]$ has modulus at most 1. Intuitively, $|e^{isX}| = 1$ for every realisation, but the complex exponential e^{isX} rotates in the complex plane as X varies; when X is widely spread, these rotations point in different directions and cancel when averaged, driving $|\phi_X(s)|$ toward zero. Concretely, if X has variance σ_X^2, then for large $|s|$ the CF decays at least as fast as $e^{-\sigma_X^2 s^2/2}$; a larger variance means faster decay.

In the PCPG integral, the CF is evaluated at the frequency $s = -uw_m$, which is proportional to the inner sample u. The relevant variance is that of the projection $w_m^T x^m$ onto the classifier direction, which equals D_m. Hence

$$\Big|\phi_{x^m|x^o}(-uw_m)\Big| \;\le\; \exp\!\Big(-\tfrac{1}{2}u^2\,D_m\Big)\,.$$

When D_m is large, this bound drops to nearly zero already for moderate $|u|$ — the integrand $h(\gamma, u)$ is effectively zero outside a narrow band $|u| \lesssim 1/\sqrt{D_m}$ around zero. Larger D_m makes this band narrower: "stronger CF damping" means the PCPG integrand is non-negligible only for very small u.

Two consequences follow from a narrow effective integration range. First, σ_A^2 decreases: the $e^{1/(8\gamma)}$ blowup that drives inner variance is large only when γ is small, which makes $u \sim \mathcal{N}(0, \gamma)$ small as well. In this regime the CF is near 1 regardless of D_m, so the blowup is not directly suppressed. However, for large γ — where most PG mass lies — the Gaussian $\mathcal{N}(0, \gamma)$ places u well outside the narrow band, and the CF kills those samples. The net effect is fewer extreme inner samples, lowering σ_A^2. Second, σ_B^2

decreases: the inner expected value $\mathbb{E}_{u|\gamma}[Z \mid \gamma]$ is near zero for any γ large enough that $\sqrt{\gamma} \gg 1/\sqrt{D_m}$ (essentially all large PG draws), and near $1/2$ only for the rare very small γ draws. As D_m grows, the threshold shifts to smaller γ and a larger fraction of PG draws gives $\mathbb{E}[Z|\gamma] \approx 0$, so the variation of this function over γ — which is σ_B^2 — shrinks. *PCPG variance decreases with the missing ratio.*

The same narrowing of the integrand reduces the Gauss–Hermite quadrature error for PCPG-GH: a more concentrated integrand is easier to integrate accurately with a fixed number of nodes, so the bias also decreases with ρ.

Effect of total dimension (fixed missing ratio ρ). As d grows with ρ fixed, $d_m = \rho d$ grows, but so does $d_o = (1-\rho)d$. Each individual weight scales roughly as $w_j = O(1/\sqrt{d})$ in a well-fit logistic regression (the discriminative signal is spread over more features), so each term in D_m is $O(1/d)$. Summing over $d_m = \rho d$ missing features gives $D_m = O(\rho)$: bounded, regardless of d.

For *MC*, the variance of $w_m^T x^m$ stays at $D_m = O(\rho)$, so the distribution of $\sigma(w^T x)$ approaches a fixed limit as $d \to \infty$ and σ_{MC}^2 neither grows nor shrinks.

For *PCPG*, D_m stays bounded, so the CF damping does not improve and the variances σ_A^2, σ_B^2 stay roughly constant.

Higher total dimension at fixed missing ratio does not improve PCPG over MC. The relative advantage of PCPG depends on the missing ratio ρ, not on the total number of features d alone.

Summary. As the missing ratio ρ grows: MC variance increases toward $1/(4N)$, while PCPG variance (both σ_A^2 and σ_B^2) decreases due to CF concentration. As total dimension d grows at fixed ρ: both estimators are largely unaffected, since D_m stays bounded. The relative advantage of PCPG over MC is therefore tied to the missing ratio, and is strongest in the high-missing-data regime where accurate marginalisation matters most.

2.5 Related Work

Here, we just focus on works that also try to calculate

$$\mathbb{E}_{P(x^m|x^o)}\left[\sigma(w^T x)\right]$$

but use a different approach.

2.5.1 Naive Conformant Learning (NaCL)

Khosravi et al. [5] first point out that this expectation is hard to compute in general: even with a naive Bayes feature distribution, it is NP-hard. They also note that the popular shortcut of mean imputation only matches the expectation when the classifier is linear *and* the features are independent — for the sigmoid it is always biased.

Their fix is to pick a feature distribution on which the expectation *is* cheap, namely a naive Bayes model over binary features. The trick is that naive Bayes and logistic regression are dual to each other: every naive Bayes induces a logistic regression, and many naive Bayes models induce the same one. They call a naive Bayes *conformant* with the given LR if both give the same class probabilities on fully observed inputs, and learn such a distribution via geometric programming ("naive conformant learning"). Once this naive Bayes is fitted, the expected prediction is just a marginal inference query and runs in linear time.

Difference to us: they keep the feature distribution simple (naive Bayes on binary inputs) so the expectation is exact and cheap; we keep the feature distribution rich (a GMM or PC on continuous inputs) and instead use a stochastic trick to handle the sigmoid. They also have to refit the feature distribution whenever the classifier changes, while ours is trained once and reused.

2.5.2 Joint Modelling via SAEM

Jiang, Josse, and Lavielle [6] assume a joint Gaussian model on the covariates, $x \sim \mathcal{N}(\mu, \Sigma)$, together with logistic regression on top. Their main contribution is on the training side: they fit (μ, Σ, β) jointly from training data that may itself contain missing values, using a stochastic-approximation EM algorithm with Metropolis–Hastings updates.

For prediction at test time, they simply do plain Monte Carlo: sample $x^{m,(s)} \sim p(x^m \mid x^o)$ from the fitted Gaussian and average,

$$p(y \mid x^o) \approx \frac{1}{S} \sum_{s=1}^{S} p\Big(y \mid x^o, x^{m,(s)}\Big)$$

This is exactly the MC baseline we compare against, just with a single Gaussian instead of a mixture as the density.

Difference to us: they sample the missing features and average the sigmoid; we never sample the missing features, but instead rewrite the sigmoid so that only the characteristic function of $p(x^m \mid x^o)$ is needed. Their feature distribution is also limited to a single Gaussian, while ours can capture multimodal shapes.

Appendix

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